Deductive concurrency verification on weak memory models

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References

Work based on:

- Lara Bargmann, Heike Wehrheim: Towards Proving Liveness on Weak Memory. to appear in: FM 2026.
- Lara Bargmann, Brijesh Dongol, Heike Wehrheim: Unifying Weak Memory Verification Using Potentials. FM (1) 2024: 519-537.
- Ori Lahav, Brijesh Dongol, Heike Wehrheim: Rely-Guarantee Reasoning for Causally Consistent Shared Memory. CAV (1) 2023: 206-229.

Motivation

Our interest:
proving correctness of concurrent programs

For example:

- concurrent queue behaving as a queue
- concurrent algorithm ensuring mutual exclusion
- software transactional memory algorithm ensuring atomic access

on a weak memory model

Message passing (MP)

Message passing example

$$\textbf{Init}\ \mathtt{x} := 0;\ \mathtt{y} := 0;$$

Thread $\mathtt{T}_1$	**Thread** $\mathtt{T}_2$
$\mathtt{x} := 5;$	$\mathtt{a} := \mathtt{y};$
$\mathtt{y} := 1;$	$\mathtt{b} := \mathtt{x};$

Shared variables: $\mathtt{x}, \mathtt{y}$
Local registers: $\mathtt{a}, \mathtt{b}$ (initially 0)

Reasoning via Hoare-style proof calculi

Objective:
Reasoning with Hoare-style assertions and proof rules

Hoare triple:

postcondition

$$\{p\} \quad C \quad \{q\}$$

precondition

program

Proof Calculus

Proof rules per programming constructs

E.g.

- Axiom of assignment

$$\{p(x := e)\}x := e\{p\}$$

- Sequential composition

$$\frac{\{p\}S_1\{r\} \qquad \{r\}S_2\{q\}}{\{p\}S_1;\ S_2\{q\}}$$

- ...

Hoare triple for MP

$$\{true\}$$

$$\textbf{Init}\ \mathtt{x} := 0;\ \mathtt{y} := 0;$$

$$\begin{array}{c|c} \mathtt{x} := 5; & \mathtt{a} := \mathtt{y}; \\ \mathtt{y} := 1; & \mathtt{b} := \mathtt{x}; \end{array}$$

$$\{\mathtt{a} = 1 \Rightarrow \mathtt{b} = 5\}\checkmark$$

Owicki-Gries proof technique for concurrent programs

Concurrency Reasoning (Owicki-Gries)

Proof outlines:

Thread $\mathtt{T}_1$	**Thread** $\mathtt{T}_2$
$\ldots$	$\{r\}$
$\{p_1\}$	$c_2;$
$c_1;$	$\ldots$
$\{p_2\}$	
$\ldots$	

Check

- local correctness
  e.g. $\{p_1\}\ c_1\ \{p_2\}$
- global correctness (non-interference)
  e.g. $\{p_2 \wedge r\}\ c_2\ \{p_2\}$

[Owicki, Gries: An axiomatic proof technique for parallel programs I]

Proof Outline for MP

$\{true\}$

Init $\mathtt{x} := 0;\ \mathtt{y} := 0;$

$\{true\}$	$\parallel$	$\{\mathtt{y} = 1 \Rightarrow \mathtt{x} = 5\}$
$\mathtt{x} := 5;$	$\parallel$	$\mathtt{a} := \mathtt{y};$
$\{\mathtt{x} = 5\}$	$\parallel$	$\{\mathtt{a} = 1 \Rightarrow \mathtt{x} = 5\}$
$\mathtt{y} := 1;$	$\parallel$	$\mathtt{b} := \mathtt{x};$
$\{true\}$	$\parallel$	$\{\mathtt{a} = 1 \Rightarrow \mathtt{b} = 5\}$

$\{\mathtt{a} = 1 \Rightarrow \mathtt{b} = 5\}$✓

Sequential Consistency (SC)

The result of any execution is the same as if the operations of all the processes were executed in some sequential order, and the operations of each individual process appear in this sequence in the order specified by its program.

Leslie Lamport, 1976

Weak memory models

Assumption for Owicki-Gries: sequential consistency (SC)

However: executions of concurrent programs on multi-core architectures do not satisfy SC

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On some memory models: $\mathtt{a} = 1 \wedge \mathtt{b} = 0$

Weak memory models

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Message passing example:

Wanted:
Reasoning technique for weak memory

$\{\mathtt{a} = 1 \Rightarrow \mathtt{b} = 5\}$ ✗

On some memory models: $\mathtt{a} = 1 \wedge \mathtt{b} = 0$

Key Idea

Instead of several proof calculi: just one

construct one proof outline

then: check whether employed proof rules are valid for specific weak memory models

Challenge

Problem:

- use operational semantics
- different memory models – different state

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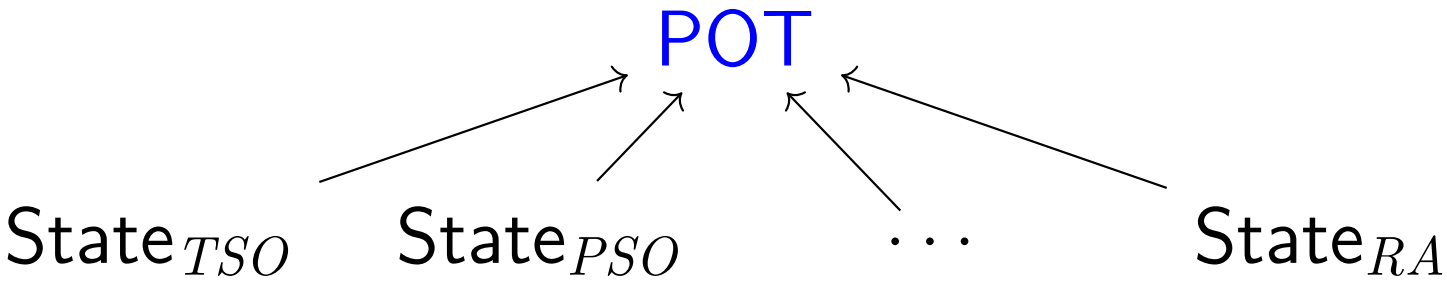
- different memory models – different state

Solution:

- potentials as unifying semantic domain
- logic and proof rules over potentials
- lifting weak memory state onto potentials



Potentials

Idea: State = potential future read options per thread

Example MP (for WM model TSO):

$\mathtt{T}_1$:

$$\begin{bmatrix} \mathtt{x} \mapsto 0 \\ \mathtt{y} \mapsto 0 \end{bmatrix}$$

$\mathtt{T}_2$:

$$\begin{bmatrix} \mathtt{x} \mapsto 0 \\ \mathtt{y} \mapsto 0 \end{bmatrix}$$

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$\mathtt{T}_2$:

$$\begin{bmatrix} \mathtt{x} \mapsto 0 \\ \mathtt{y} \mapsto 0 \end{bmatrix}$$

$$\mathtt{T}_1 \mapsto \mathtt{x} := 5$$

$$\begin{bmatrix} \mathtt{x} \mapsto 5 \\ \mathtt{y} \mapsto 0 \end{bmatrix}$$

$$\begin{bmatrix} \mathtt{x} \mapsto 0 \\ \mathtt{y} \mapsto 0 \end{bmatrix} \cdot \begin{bmatrix} \mathtt{x} \mapsto 5 \\ \mathtt{y} \mapsto 0 \end{bmatrix}$$

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$$\mathtt{T}_1 \mapsto \mathrm{y} := 1$$

$$\begin{bmatrix} \mathrm{x} \mapsto 5 \\ \mathrm{y} \mapsto 1 \end{bmatrix}$$

$$\begin{bmatrix} \mathrm{x} \mapsto 0 \\ \mathrm{y} \mapsto 0 \end{bmatrix} \cdot \begin{bmatrix} \mathrm{x} \mapsto 5 \\ \mathrm{y} \mapsto 0 \end{bmatrix} \cdot \begin{bmatrix} \mathrm{x} \mapsto 5 \\ \mathrm{y} \mapsto 1 \end{bmatrix}$$

Assertions for Sets of States

Logic for assertions:

Piccolo (Potential based causal consistency logic)

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State of $\mathtt{T}_2$:

$$\begin{bmatrix} \mathrm{x} \mapsto 0 \\ \mathrm{y} \mapsto 0 \end{bmatrix} \cdot \begin{bmatrix} \mathrm{x} \mapsto 5 \\ \mathrm{y} \mapsto 0 \end{bmatrix} \cdot \begin{bmatrix} \mathrm{x} \mapsto 5 \\ \mathrm{y} \mapsto 1 \end{bmatrix}$$

State of $\mathtt{T}_1$:

$$\begin{bmatrix} \mathrm{x} \mapsto 5 \\ \mathrm{y} \mapsto 1 \end{bmatrix}$$

Example assertions:

$$\mathtt{T}_2 \ltimes [\mathrm{y} \neq 1] \; ; \; [\mathrm{x} = 5]$$

(intervals)

$$\mathtt{T}_1 \uparrow \mathrm{x}$$

(view maximality)

Proof Rules of Piccolo

Basically:

- new axioms of Assignment
- rest: kept from Owicki-Gries

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Three example axioms:

$$\{\tau \uparrow x\} \quad \tau \mapsto x := e \quad \{\tau \uparrow x \wedge \tau \ltimes [x = e]\}$$

$$\{\tau \ltimes I_\tau \wedge \pi \ltimes I \; ; \; I_\tau\} \quad \tau \mapsto x := e \quad \{\pi \ltimes I \; ; \; I_\tau\}$$

$$(x \notin fv(I_\tau), \tau \neq \pi)$$

$$\{\tau \ltimes [e(a := x)]\} \quad \tau \mapsto a := x \quad \{e\}$$

Proof Outline of MP in Piccolo

$\{true\}$

Init $\mathtt{x} := 0;\ \mathtt{y} := 0;$

Thread $\mathtt{T}_1$	**Thread** $\mathtt{T}_2$
$\{\mathtt{T}_1 \uparrow \mathtt{x}\}$	$\{\mathtt{T}_2 \ltimes [\mathtt{y} \neq 1]\ ;\ [\mathtt{x} = 5]\}$
$\mathtt{x} := 5;$	$\mathtt{a} := \mathtt{y};$
$\{\mathtt{T}_1 \ltimes [\mathtt{x} = 5]\}$	$\{\mathtt{a} = 1 \Rightarrow \mathtt{T}_2 \ltimes [\mathtt{x} = 5]\}$
$\mathtt{y} := 1;$	$\mathtt{b} := \mathtt{x};$
$\{true\}$	$\{\mathtt{a} = 1 \Rightarrow \mathtt{b} = 5\}$

$\{\mathtt{a} = 1 \Rightarrow \mathtt{b} = 5\}$✓

Soundness of Rules wrt. WM models

Use liftings of WM state to potentials

$$
\begin{array}{ccc}
\{p\} & cmd & \{q\} \\
\rotatebox{90}{$\models$} & & \rotatebox{90}{$\models$} \\
state_{POT} & & state'_{POT} \\
\uparrow & & \uparrow \\
s_{WM} & \xrightarrow{cmd}_{WM} & s'_{WM}
\end{array}
$$

Stability

For safety proofs:
stability of logic under memory model internal steps

E.g.,

$$\begin{bmatrix} \mathrm{x} \mapsto 0 \\ \mathrm{y} \mapsto 0 \end{bmatrix} \cdot \begin{bmatrix} \mathrm{x} \mapsto 5 \\ \mathrm{y} \mapsto 0 \end{bmatrix} \cdot \begin{bmatrix} \mathrm{x} \mapsto 5 \\ \mathrm{y} \mapsto 1 \end{bmatrix} \xrightarrow{\iota} \begin{bmatrix} \mathrm{x} \mapsto 5 \\ \mathrm{y} \mapsto 0 \end{bmatrix} \cdot \begin{bmatrix} \mathrm{x} \mapsto 5 \\ \mathrm{y} \mapsto 1 \end{bmatrix}$$

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i.e., we like to guarantee the following for weak memory states s, s', Piccolo formulae φ and memory steps ι

$$s \models \varphi \wedge s \xrightarrow{\iota}_{WM} s' \Rightarrow s' \models \varphi$$

Check Proof Outline for WM model

Proof Outline of MP in Piccolo:

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$\mathtt{x} := 5;$	$\mathtt{a} := \mathtt{y};$
$\{\mathtt{T}_1 \ltimes [\mathtt{x} = 5]\}$	$\{\mathtt{a} = 1 \Rightarrow \mathtt{T}_2 \ltimes [\mathtt{x} = 5]\}$
$\mathtt{y} := 1;$	$\mathtt{b} := \mathtt{x};$
$\{true\}$	$\{\mathtt{a} = 1 \Rightarrow \mathtt{b} = 5\}$

$\{\mathtt{a} = 1 \Rightarrow \mathtt{b} = 5\}$✓

Check whether all employed proof rules are sound under WM model

Proof valid under SC, TSO, RA, SRA but not PSO

Summary

One proof – valid under several memory models

Current work: liveness